

EOL

MES + Camera CMS

User Manual

Toyota Boshoku Device India

Bawal Plant | YNC SS Line

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What's inside

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Purpose & Methodology

Why we built this and how the pieces fit

2

Collector → 4 Departments

One poll, four views — the heart of the system

3

End-to-End Architecture

PLC bit → database → screen, in seconds

4

Network & IPs

Where each component physically lives

5

Directory Structure

Where every important file is kept

6

Page → Data → DB Map

Which screen reads which table

7

How to Start / Roles

One launcher + per-role access

1. Purpose & Methodology

Why this project exists

VISIBILITY

Operators on the floor and managers in the office see the SAME numbers, in real time, with no manual entry.

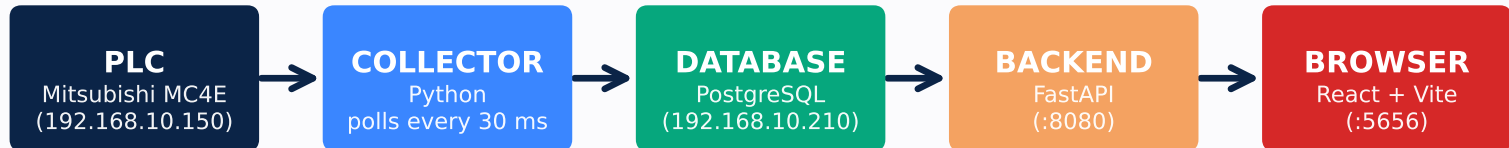
TRACEABILITY

Every cycle is recorded on video. Click a defect in the dashboard → instantly play that exact moment.

AUTOMATION

PLC bits drive everything. No human in the loop between the machine and the screen.

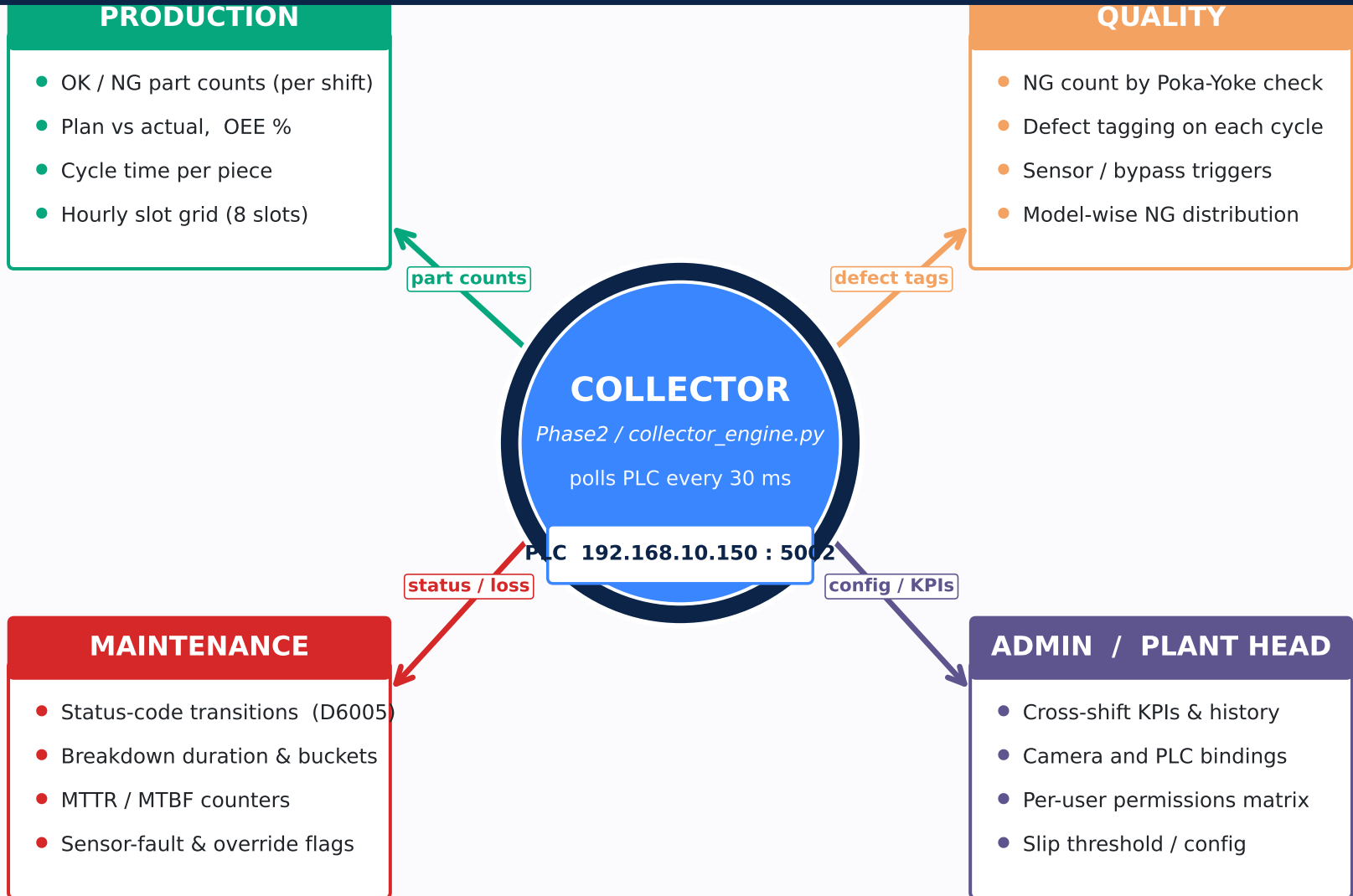
How it works



End-to-end latency: ~1-2 seconds. All on one server PC for now (192.168.10.185).

2. One Collector → Four Departments

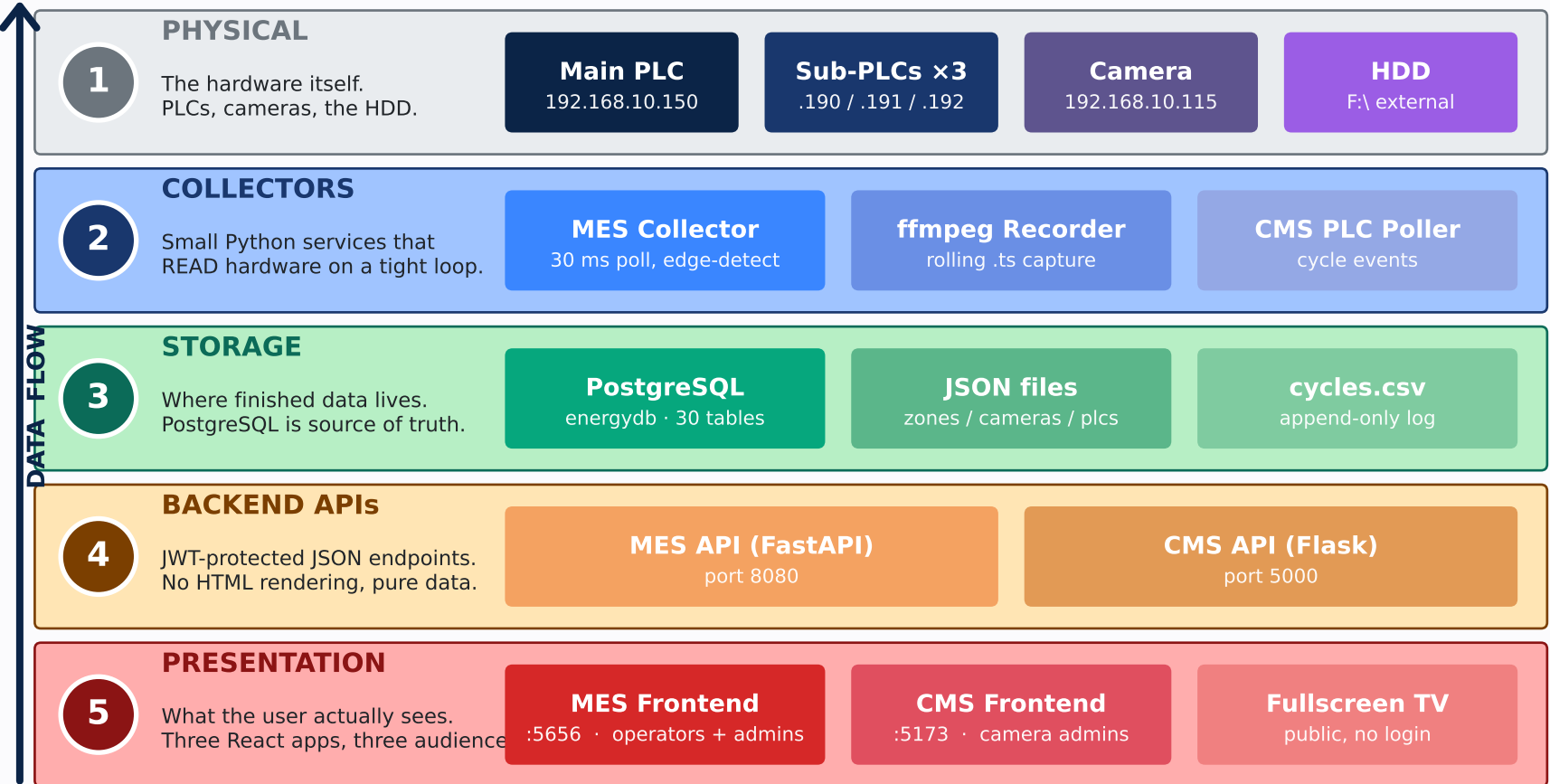
Same PLC data, four different views



ONE poll loop is the source of truth. Every department reads the same numbers — no spreadsheet stitching, no manual entry.

3. End-to-End Architecture

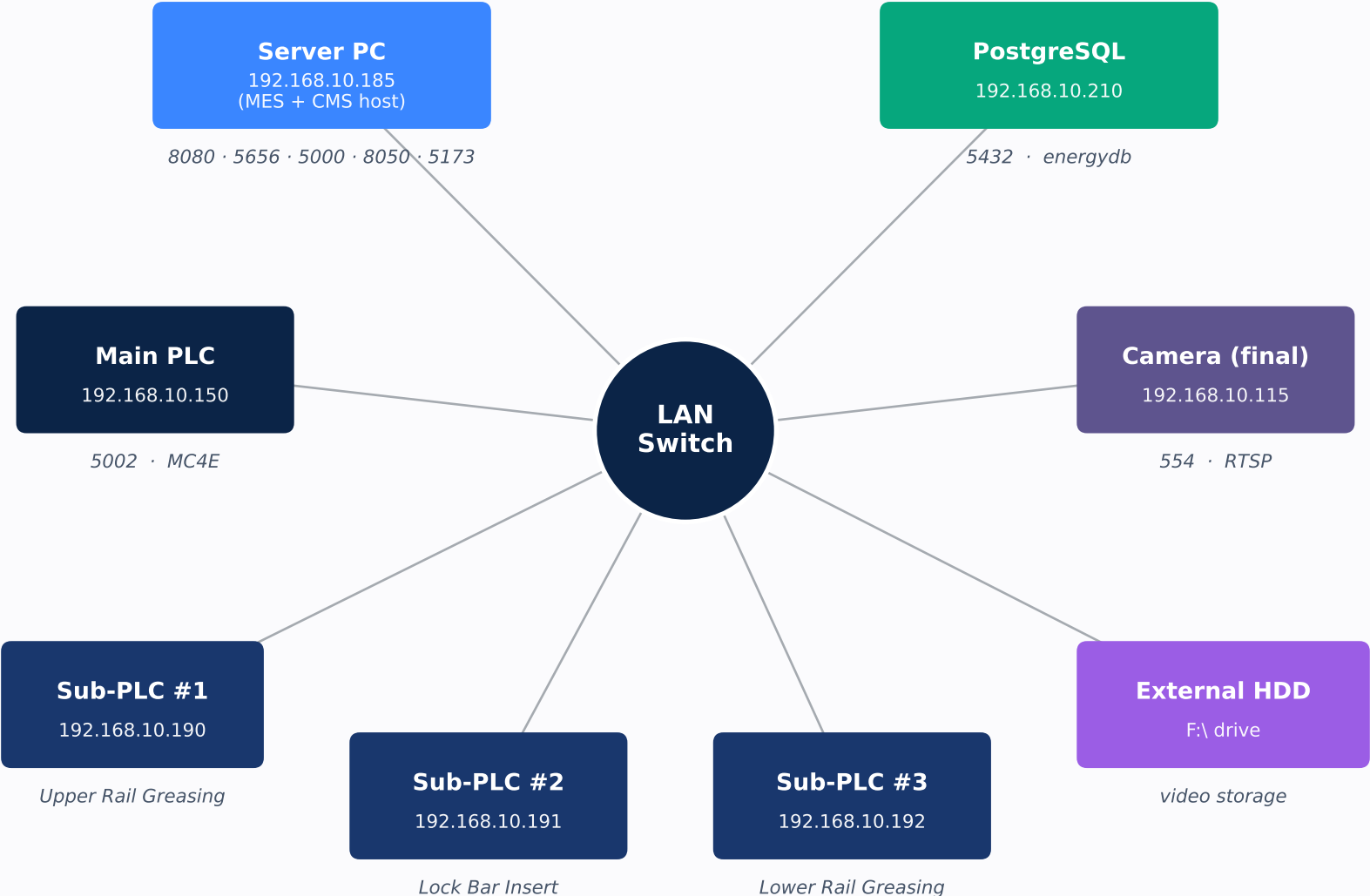
5 stacked layers · data flows bottom → top



Every screen number can be traced down to one PLC bit at the bottom. No layer skips another.

4. Network & IP Map

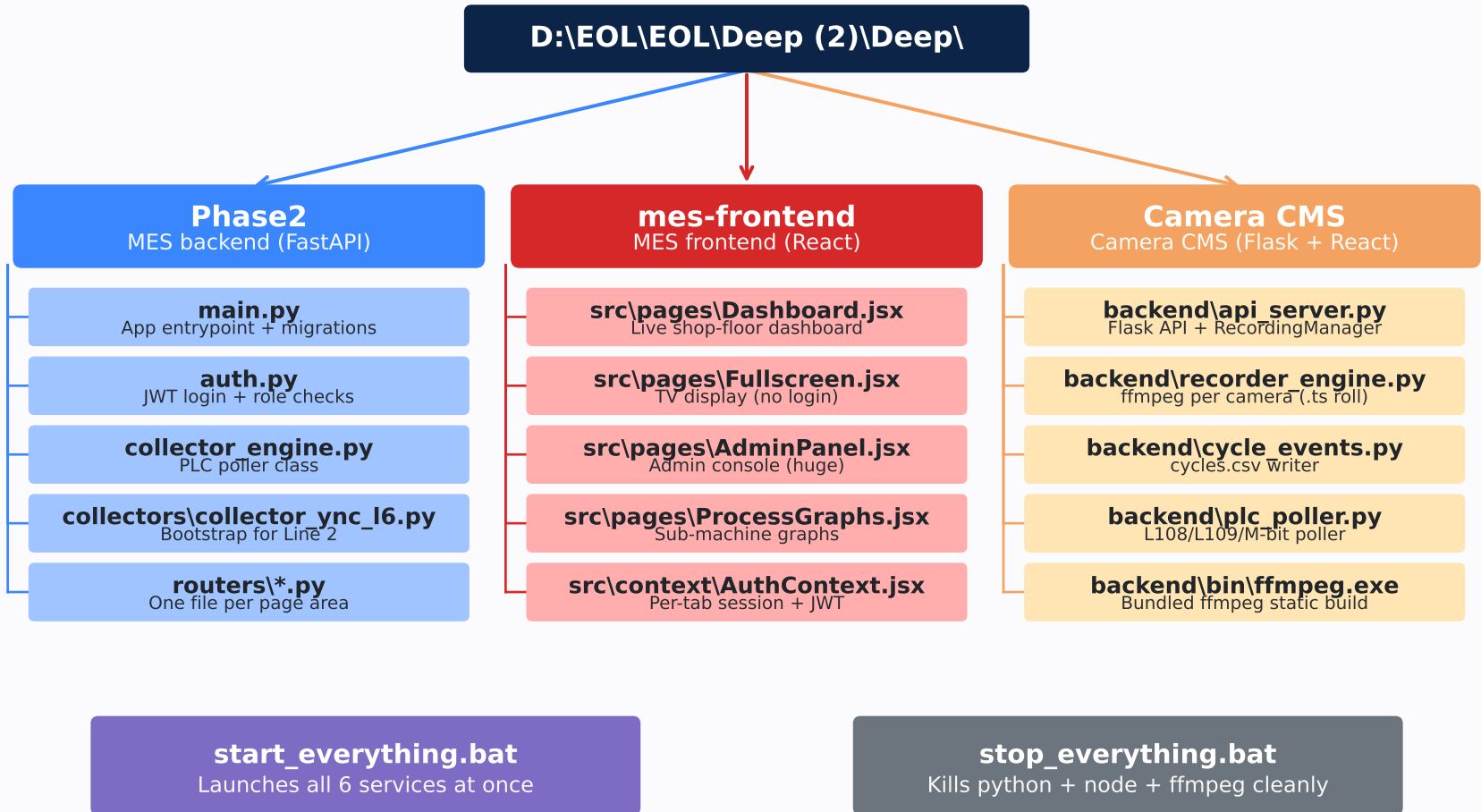
All hosts on the factory LAN



Storage path is configurable from the CMS UI: Configuration → System Settings.

5. Directory Structure

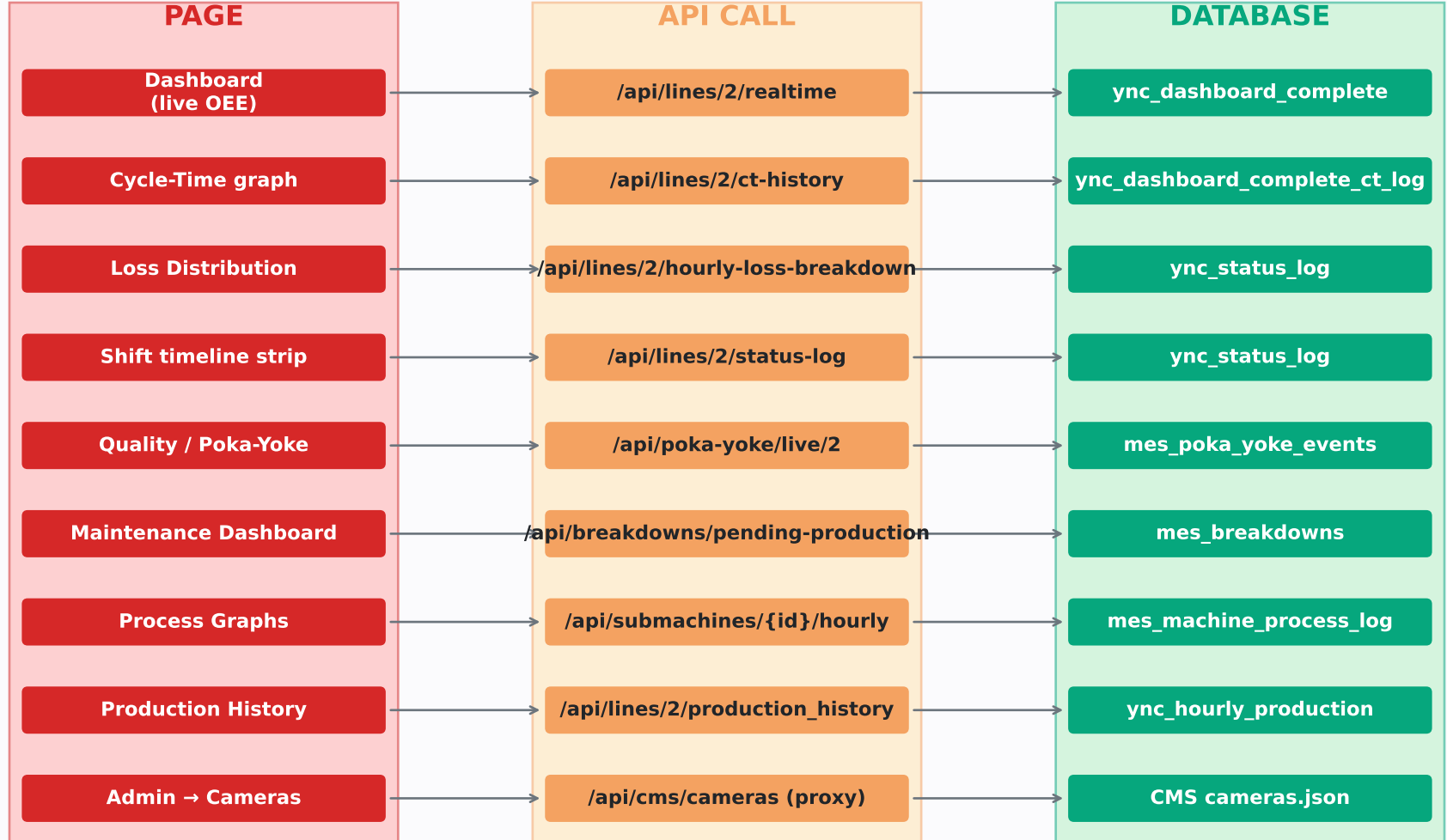
What lives where



Tree shows only the files you'll touch. Hidden files (caches, node_modules) live alongside but you should never edit them.

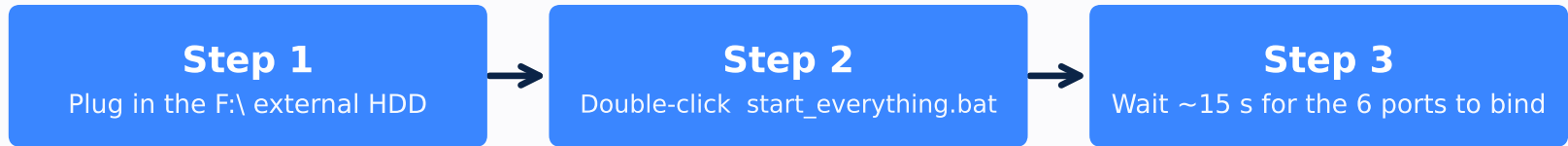
6. Page → Data → Database

Which screen reads which table

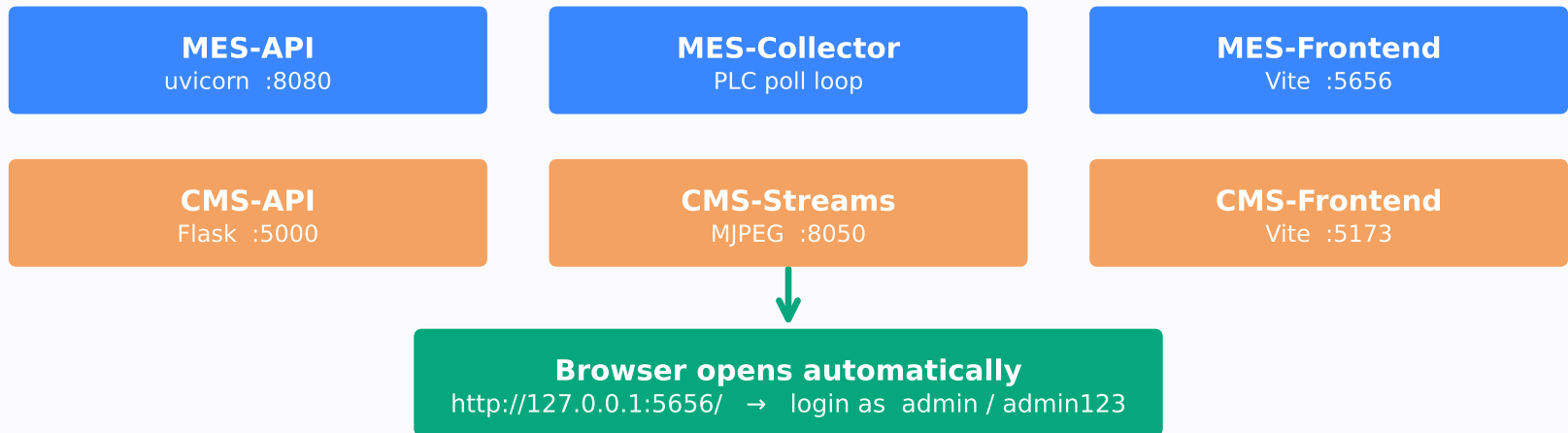


7. How to Start the System

One double-click → six services



What opens automatically



To stop everything: double-click stop_everything.bat

If a single service hangs, just close its window — the others keep running. Collector lock auto-releases after 30 s.

7. User Roles & Colour Scheme

Who sees what

ADMIN

admin

Full control — every page, every action.
Users, Lines, Plants, Cameras, Slip threshold.



PRODUCTION

production

Live dashboard, hourly grid, breakdown raise (their half), cycle-time history.



MAINTENANCE

maintenance

Breakdown dashboard, MTTR / MTBF tiles, CAPA, fill maintenance half of slips.



QUALITY

quality

Poka-yoke live, Quality deviations, NG counter, PY master assignments.



OPERATOR

operator

Read-only dashboard, cycle-time view.
No admin, no editing.



PLANT HEAD

plant_head

Same as Admin — high-level oversight.



Per-user override

Admin can fine-tune access via Admin Panel → Users → Permissions Matrix (none / read / full × every page). Overrides win over the role defaults above.

7. Page Access Map

Heat-map view · one glance = one answer

	ADMIN	PLANT HD	PROD	MAINT	QUAL	OP	DEPT
Login	●	●	●	●	●	●	●
Dashboard	●	●	●	●	●	●	●
Fullscreen TV	◇	◇	◇	◇	◇	◇	◇
Cycle-Time History	●	●	●	●	●	◐	●
Loss Distribution	●	●	●	●	●	●	●
Hourly Production Grid	●	●	●	●	●	◐	●
Process Graphs	●	●	●	●	●	◐	●
Quality / Poka-Yoke	●	●	◐	◐	●	◐	●
Maintenance Dashboard	●	●	◐	●	◐	◐	●
Breakdown raise / fill	●	●	P	M	▪	▪	▪
CAPA / Deviations	●	●	▪	M	Q	▪	Q
Admin → Users	●	●	▪	▪	▪	▪	▪
Admin → Lines / Plants	●	●	▪	▪	▪	▪	▪
Admin → Cameras	●	●	▪	▪	▪	▪	▪
Admin → Slip Threshold	●	●	▪	▪	▪	▪	▪
AI Assistant chat	●	●	●	●	●	▪	●
Camera CMS portal	●	●	▪	▪	▪	▪	▪

● Full

◐ Read-only

▪ No access

◇ Public

P Prod half

M Maint half

Q Qual only